

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE 2004

Mathematics Extension 2

TIME ALLOWED: 3 hours

Instructions:

- Write your name and class at the top of this page, and at the top of each answer sheet.
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions are of equal value and may be attempted.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.

(FOR MARKERS USE ONLY)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL

QUESTION 1:

- 2 (a) (i) $\int \frac{dx}{x^2 + 6x + 13}$
- 2 (ii) $\int_0^4 \frac{dx}{(2x+1)\sqrt{2x+1}}$
- 3 (b) Show that $\int x^n e^x dx = x^n e^x - n \int x^{n-1} e^x dx$
Hence find $\int x^2 e^x dx$
- 3 (c) Sketch the locus of the point Z in the Argand plane, which moves so that $\arg(z-1) = \frac{\pi}{2}$
- 3 (d) (i) Find values of A, B and C so that $\frac{13}{(x^2+4)(x+3)} \equiv \frac{Ax+B}{x^2+4} + \frac{C}{x+3}$
- 2 (ii) Hence find $\int \frac{13dx}{(x^2+4)(x+3)}$

QUESTION 2:

- (a) If $z = 1 + \sqrt{3}i$ find
- 4 (i) $\bar{z}$ (ii) $|z|$ (iii) $\arg z$ (iv) $\arg(iz)$
- (b) Express $z = 1 + \sqrt{3}i$ in mod-arg form and hence find
- 3 (i) $\sqrt{z}$ (ii) z^6 (in simplest form)
- 3 (c) On the same set of axes, sketch $y = |x-1|$ and $y = |x+1|$ and then use this graph, or otherwise, to find the value of k if $|x-1| + |x+1| \geq k$ for all values of x.
- 3 (d) (i) The point W represents the complex number $w = a + ib$, and $w = \frac{z}{z+1}$ where $z = x + iy$.
The point Z representing the complex number z moves along the y-axis only.
Show that $a = \frac{y^2}{1+y^2}$ and $b = \frac{y}{1+y^2}$
- 2 (ii) Find the locus of W both algebraically and geometrically.

QUESTION 3 :

- 4 (a) (i) Using the method of Mathematical Induction, prove deMoivre's Theorem

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

- 2 (ii) Prove that the points on the Argand Diagram with co-ordinates representing $(\operatorname{cis} \frac{\pi}{3})^n$ for $n=1,2,3,4,5,6$ are the vertices of a regular hexagon inscribed in a circle of radius 1 unit.

- (b) If $f(x) = \cos x + i \sin x$,

- 1 (i) Find $f(0)$

- 1 (ii) Show that $\frac{f'(x)}{f(x)} = i$

- 3 (iii) By integrating both sides of part (ii), deduce that $\cos x + i \sin x = e^{ix}$.
This is EULER's THEOREM.

- 1 (iv) Using Euler's Theorem from part (iii), prove deMoivre's Theorem

- 1 (c) (i) Describe the locus of the point z , where $|z - a| = r$

- 2 (ii) If $|z - a| = r$ and $|z - b| = s$ what is the geometric significance when $|a - b| = r + s$

QUESTION 4:

3 (a) By using t-results, or otherwise, find $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{d\theta}{\sin \theta}$ leaving your answer in exact form

(b) Sketch the following on different sets of axes showing all important features

(DO NOT USE CALCULUS)

8

(i) $y = \frac{|x|}{x}$

(ii) $y = \ln\left(\frac{1}{x^2}\right)$

(iii) $y = |\tan x|$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

(iv) $y = \sec x$ for $-2\pi \leq x \leq 2\pi$

4 (c) Use calculus, or otherwise, to sketch $y = \frac{e^x}{x}$ showing all stationary points and asymptotes, if they exist.

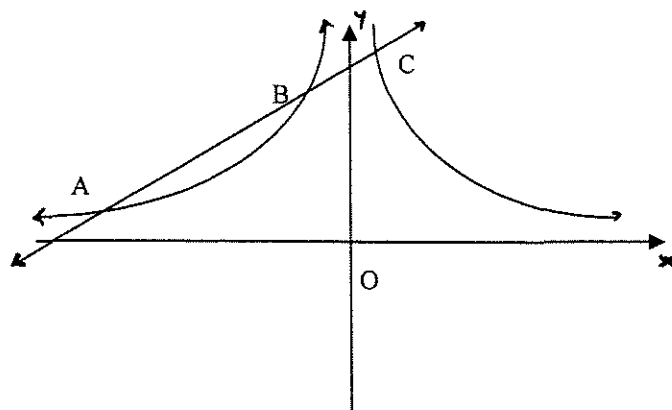
QUESTION 5:

(a) For the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ find

- 3 (i) the eccentricity
(ii) the co-ordinates of the foci
(iii) the equations of the directrices

3 (b) Given that $P(x) = 3x^3 - 11x^2 + 8x + 4$ has a double root, fully factorise $P(x)$

(c)



In the diagram above, the points A, B and C represent the points of intersection of the line $y = 4x + 8$ and the curve $y = \frac{1}{x^2}$. The x-values of A, B and C are α , β and γ

1 (i) Show that α , β and γ satisfy $4x^3 + 8x^2 - 1 = 0$

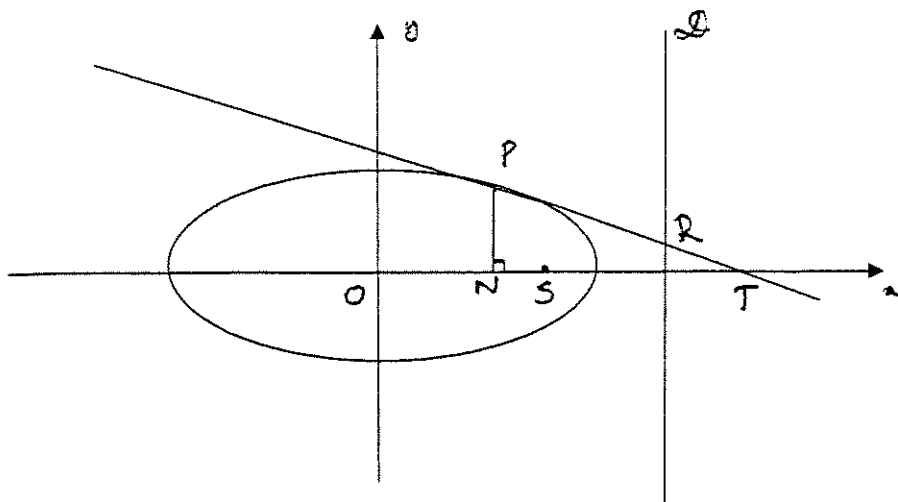
3 (ii) Find a polynomial with roots α^2 , β^2 and γ^2

2 (iii) Find $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$

3 (iv) Prove that $OA^2 + OB^2 + OC^2 = 132$ where O is the origin.

QUESTION 6:

(a)



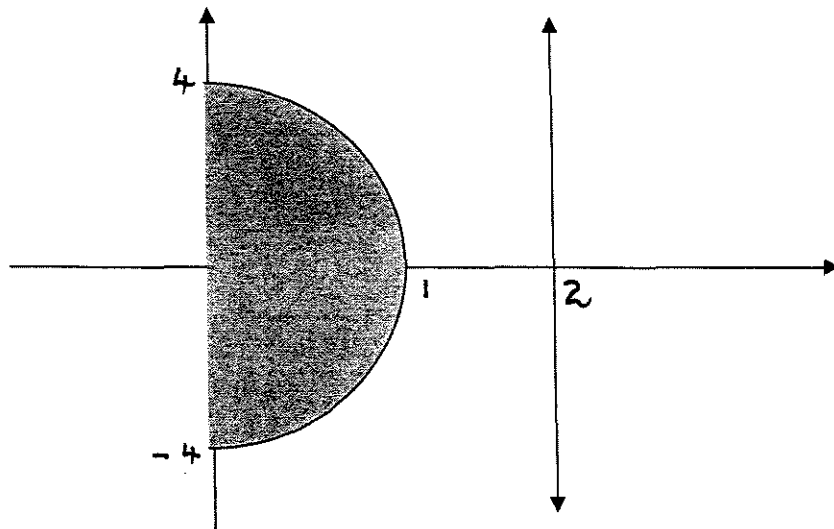
$P(a\cos\theta, b\sin\theta)$ is any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The tangent at P cuts the major axis in T and the Directrix in R. N is the foot of the perpendicular from P to the major axis, O is the centre and S is the focus.

- 2 (i) Show that the equation of the tangent at P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$
- 2 (ii) Show that $ON \cdot OT = a^2$
- 5 (iii) Showing all steps carefully, prove that $\angle PSR = 90^\circ$

QUESTION 6 continues on the next page...

QUESTION 6 continued...

(b)



A solid S is formed by rotating the region bounded by the parabola $y^2 = 16(1-x)$ and the y -axis through 2π about the line $x=2$.

- 3 (i) Use the method of cylindrical shells to show that the volume of S is given by

$$\int_0^1 16\pi(2-x)\sqrt{1-x} dx$$

- 3 (ii) Calculate this definite integral by using the substitution $u=1-x$ (or otherwise)

QUESTION 7:

(a) If 1, w_1 and w_2 are the cube roots of unity, prove that

2 (i) $w_1 = \overline{w_2} = w_2^2$

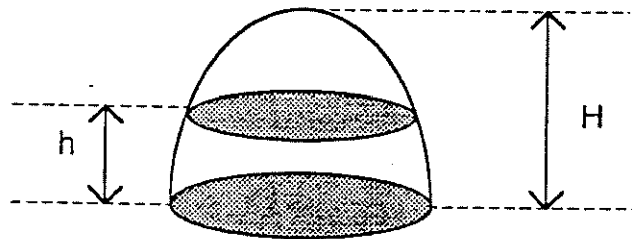
1 (ii) $w_1 + w_2 = -1$

1 (iii) $w_1 w_2 = 1$

3 (b) (i) By using the substitution $x = a \sin \theta$ or otherwise, verify that $\int_0^a \sqrt{a^2 - x^2} dx = \frac{1}{4} \pi a^2$

2 (ii) Deduce that the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab

6 (iii)



The diagram above shows a mound of height H . At height h above the horizontal base, the horizontal cross section of the mound is elliptical in shape, with equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \lambda^2 \quad \text{where } \lambda = 1 - \frac{h^2}{H^2}$$

and x and y are co-ordinates in the plane of cross section.

Show that the volume of the mound is $\frac{8\pi abH}{15}$

QUESTION 8:

- 2 (a) (i) Show that the equation of the tangent to the Hyperbola $xy = c^2$ at the point $P(cp, \frac{c}{p})$ is $x + p^2y = 2cp$

- 2 (ii) If the tangents at the points P and $Q(cq, \frac{c}{q})$ meet at the point $R(x_1, y_1)$ prove that

$$(\alpha) \quad pq = \frac{x_1}{y_1}$$

and that $(\beta) \quad p + q = \frac{2c}{y_1}$

- 2 (iii) If the length of the chord PQ is d units, show that

$$d^2 = c^2(p - q)^2 \left\{ 1 + \frac{1}{p^2 q^2} \right\}$$

- 3 (iv) Further, if d in part (iii) above remains constant, deduce that the locus of R is given by

$$4c^2(x^2 + y^2)(c^2 - xy) = x^2 y^2 d^2$$

- (b) If n is a positive integer and $f(x) = e^{-x}(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!})$, $x \geq 0$

- 3 (i) Show that $f(x)$ is a decreasing function

NOTE: $n! = 1 \times 2 \times 3 \times 4 \dots (n-1)n$

- 3 (ii) Deduce that for $x > 0$ and n any positive integer,

$$e^x \geq 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$$

End of Examination

THSC MATHS EXT 2. 2004 SOLUTIONS.

Question 1:

$$(a)(i) \int \frac{dx}{(x+3)^2+4} = \frac{1}{2} \tan^{-1}\left(\frac{x+3}{2}\right) + k$$

$$(ii) \int_0^4 (2x+1)^{-3/2} dx = -\left(2x+1\right)^{-1/2} \Big|_0^4$$

$$= -9^{-1/2} + 1^{-1/2}$$

$$= \frac{2}{3}$$

$$(b) \int x^2 e^x dx = \int x^2 \frac{d}{dx} e^x dx$$

$$= e^x x^2 - \int e^x \cdot 2x dx$$

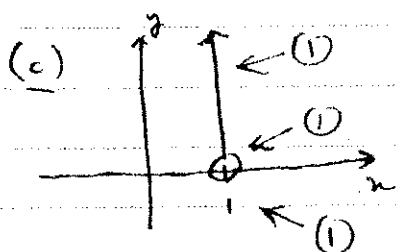
$$n=2 \therefore \int x^2 e^x dx = e^x x^2 - 2 \int e^x x dx$$

$$\int_0^1 x e^x dx = e^x x - \int e^x dx$$

$$= e^x x - e^x$$

$$\therefore \int x^2 e^x dx = e^x x^2 - 2(e^x x - e^x)$$

$$= e^x x^2 - 2e^x x + 2e^x$$



$$(d)(i) \frac{13}{(x^2+4)(x+3)} = \frac{Ax+B}{x^2+4} + \frac{C}{x+3}$$

$$\therefore (Ax+B)(x+3) + C(x^2+4) = 13$$

$$\therefore A + C = 0$$

$$3A + B = 0$$

$$3B + 4C = 13$$

$$\text{Now } \left. \begin{array}{l} C = 1 \\ B = 3 \\ A = -1 \end{array} \right\} \Rightarrow \frac{Exp^x}{x^2+4} + \frac{1}{x+3}$$

① (d) (ii)

$$\int \frac{13dn}{(n^2+4)(n+3)} = \int \frac{dn}{n+3} + \int \frac{3-n}{n^2+4} \frac{1}{2} dn$$

$$= \ln(n+3) + \frac{3}{2} \ln|n| - \frac{1}{2} \ln(n^2+4)$$

① ①

EXAMINER'S COMMENT:

- Average mark was about 12/15. Generally strongly done.
- SPECIAL COMMENTS:

1 (a)(ii) People forgot about the 2 coming out of the bracket when differentiated:

es $\frac{d}{dx} (2x+1)^{-1/2} = -\frac{1}{2} \textcircled{2} (2x+1)^{-3/2}$

(b) Not a big error, but the formula could have been used 2 times to evaluate $\int x^2 e^x dx$ and $\int e^x dx$.

most common ERROR: The question required you to show the result, not just quote the result because you knew it was true. This was the most common.

" $u = x^n$ $u' = nx^{n-1}$ $v = e^x$ $v' = e^x$

then quoting the question! This leaves me to work out the in-between part - what was your job!!

(d) Trouble was with the $\int \frac{3-n}{n^2+4}$ part.

And the negatives involved!

QUESTION 2

(a) $z = 1 + \sqrt{3}i$

(i) $\bar{z} = 1 - \sqrt{3}i$

(ii) $|z| = \sqrt{1+3}$
 $= 2$

(iii) $\arg z = \tan^{-1} \frac{\sqrt{3}}{1}$
 $= \pi/3$

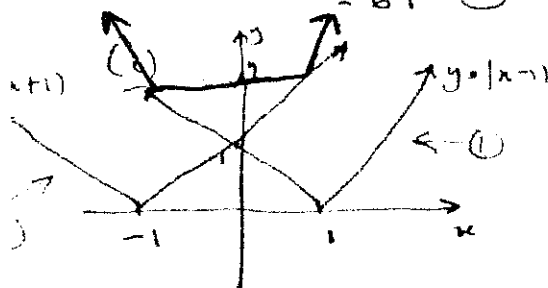
(iv) $\arg iz = \arg(-\sqrt{3} + i)$
 $= \tan^{-1}(-\frac{1}{\sqrt{3}})$
 $= 5\pi/6$

$0 \leq \arg z < 2\pi$ just $\theta = \pi/3 + \pi/2$
 $= 5\pi/6$

(b) $z = 2 \operatorname{cis} \pi/3$ (1)

$\therefore$ (i) $\sqrt{z} = \sqrt{2} \operatorname{cis} \pi/6 \left(\frac{\sqrt{6}}{2} + \frac{\sqrt{2}i}{2} \right)$

(ii) $z^6 = 2^6 \operatorname{cis} 2\pi$
 $= 64$ (1)



$k = 2$ (1)

(d) $w = \frac{z}{z+1} = a+ib$

$\therefore a+ib = \frac{x+iy}{x+iy+1}$

Now since z moves along the y -axis

$x = 0$ (1)

$\therefore a+ib = \frac{iy}{iy+1} \rightarrow \times \frac{1-iy}{1-iy}$
 $= \frac{iy(1-iy)}{1+y^2}$
 $= \frac{iy+y^2}{1+y^2}$ (1)

$\therefore a = \frac{y^2}{1+y^2}$ and $b = \frac{y}{1+y^2}$ (1)

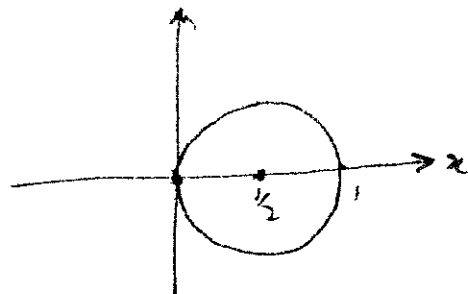
(ii) $y=0 \quad a=b=0$

$y \rightarrow \infty \quad a \rightarrow 1$

$y \rightarrow \pm\infty \quad b \rightarrow 0$

$y=1 \quad a=1/2 \quad b=1/2$

$y=-1 \quad a=1/2 \quad b=-1/2$



circle centre $(\frac{1}{2}, 0)$, $r = \frac{1}{2}$

$a = \frac{y^2}{1+y^2}$ $b = \frac{y}{1+y^2}$
 $a^2 = \frac{y^4}{(1+y^2)^2}$ $b^2 = \frac{y^2}{(1+y^2)^2}$
 $= \frac{y^2(1+y^2)}{(1+y^2)^2} - \frac{y^2}{(1+y^2)^2}$
 $= \frac{y^2}{(1+y^2)} - b^2$
 $= a - b^2$

$\therefore a^2 - a + b^2 = 0$

$\therefore (a - \frac{1}{2})^2 + b^2 = 0 + 1/4$

circle, centre $(\frac{1}{2}, 0)$ $r = \frac{1}{2}$

QUESTION 3

4(a) (i) For $n=1$, $(\cos \theta + i \sin \theta)^1 = \cos 1\theta + i \sin 1\theta$
 $\therefore$ True for $n=1$ (1)

Assume the formula is true for $n=k$,

ie. $(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$

For $n=k+1$

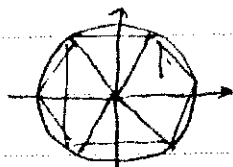
$$\begin{aligned} (\cos \theta + i \sin \theta)^{k+1} &= (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta) \quad (2) \\ &= (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta) \\ &= \cos \theta \cos k\theta - \sin \theta \sin k\theta + i(\cos \theta \sin k\theta + \sin \theta \cos k\theta) \\ &= \cos(k+1)\theta + i \sin(k+1)\theta \end{aligned}$$

$\therefore$ If the formula is true for $n=k$, it is true for $n=k+1$

But it is true for $n=1$, $\therefore$ it is true for $n=2$ and so on (1)

ie. true $\forall n$.

(ii) The points are $\text{cis } \frac{\pi}{3}$, $\text{cis } \frac{2\pi}{3}$, $\text{cis } \pi$, $\text{cis } \frac{4\pi}{3}$, $\text{cis } \frac{5\pi}{3}$, $\text{cis } 2\pi$ (1)
 which are 6 points equally-spaced about the unit circle.



So cut off equal chords (1)

$\therefore$ form a regular hexagon

(b) $f(z) = \cos z + i \sin z$

(i) $f(0) = 1$ (1)

(ii) $f'(z) = -\sin z + i \cos z$
 $= i(\cos z + i \sin z)$
 $= i f(z)$

$\therefore f'(z)/f(z) = i$ (1)

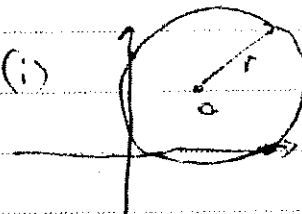
(iii) $\int \frac{f'(z)}{f(z)} dz = \int i dz$

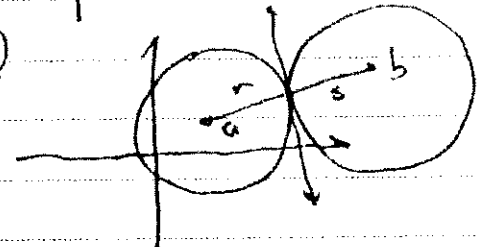
$\therefore \log_e f(z) = iz + c$ (1)

At $z=0$, $\log_e(1) = c \Rightarrow c=0$ (1)

$\therefore \log_e f(z) = iz \Rightarrow f(z) = e^{iz}$ (1)

Q 3 (b) (iv) $\cos x + i \sin x = e^{ix}$
 $\therefore (\cos x + i \sin x)^n = e^{inx} = e^{i(nx)}$ (1)
 $= \cos nx + i \sin nx$ (1)

(c) (i)  A circle of radius r , centre the point a . (1)

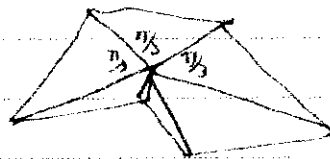
(ii)  Two circles, centres a and b which share a common tangent (ie just touch). (1)

EXAMINERS COMMENTS:

(a) Unless you don't know how to progress from one step to another, don't speculate - it looks like a fudge!

And, after all, the answer is already there! - Induction is a very formal method. In general reasonably done!

(c) (ii) The pertinent points here are that the angles are equal and that the moduli are all 1, i.e. evenly spaced about a unit circle. [↑] This was frequently omitted, and without it the diagram could have been



(b) (iii) 75% of the candidates left off the $+k$ when integrating. To evaluate the k you needed your information from part (i).

(c) (i) Well done!

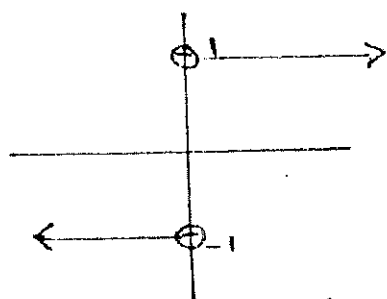
(ii) Carefully done!

QUESTION 4

a(a) $\int_{\pi/3}^{\pi/2} \frac{d\theta}{\sin\theta} = \int_{1/\sqrt{3}}^1 \frac{2dt}{1+t^2} \cdot \frac{1+t^2}{2t}$ by t -result
 $= \int_{1/\sqrt{3}}^1 \frac{dt}{t}$
 $= [\ln t]_{1/\sqrt{3}}^1$
 $= -\ln\sqrt{3} \text{ or } \frac{1}{2}\ln 3$

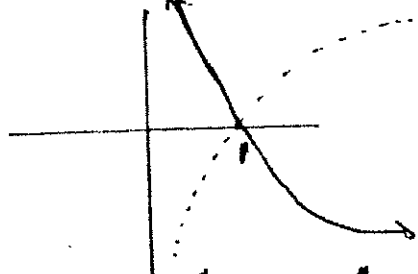
b)

i)

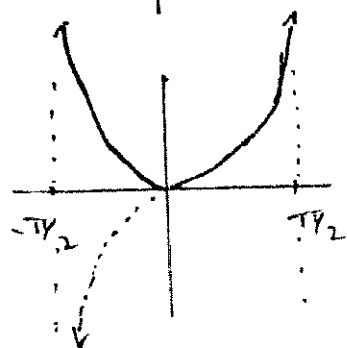


ii)

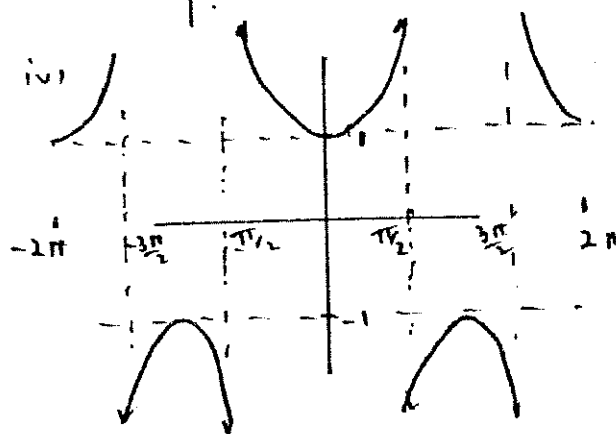
$$y = -2 \ln x$$



iii)



iv)



c) $y = \frac{e^x}{x}$

$$\frac{dy}{dx} = \frac{x \cdot e^x - e^x}{x^2}$$

$$\frac{dy}{dx} = 0 \quad x = 1$$

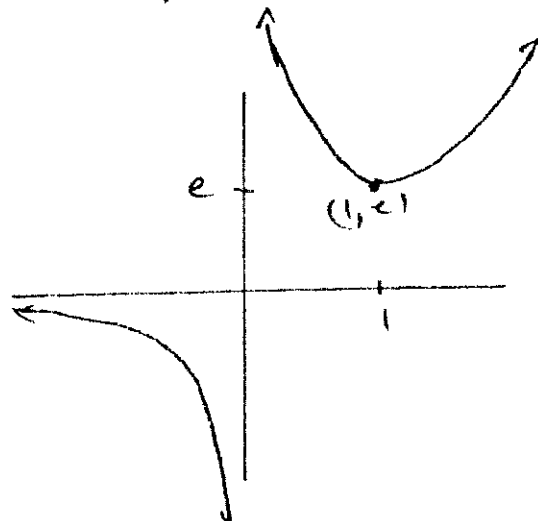
$$x = 1, y = e$$

undefined at $x = 0$

$$x \rightarrow 0^+ \quad y \rightarrow \infty$$

$$x \rightarrow 0^- \quad y \rightarrow -\infty$$

$$x \rightarrow -\infty \quad y \rightarrow 0$$



Question 5:

(a) $\frac{x^2}{a} + \frac{y^2}{4} = 1 \Rightarrow a=3, b=2$

(i) $e = \pm \sqrt{5}/3$ (1)

(ii) $S: (\pm\sqrt{5}, 0)$ (1)

(iii) $\Delta: x = \pm \frac{2}{\sqrt{5}}$ (1)

(b) $P'(x) = 9x^2 - 22x + 8$

For Double Root, $P'(x) = 0$

$\therefore (9x - 4)(x - 2) = 0$

$\therefore x = 4/9$ or $x = 2$ (1)

$P(2) = 0 \Rightarrow x = 2$ is the double root (1)

By inspection $P(x) = (x-2)^2(3x+1)$

(c) (i) $y = 4x + 8 = \frac{1}{x^2}$

$\therefore 4x^3 + 8x^2 - 1 = 0$ solves both equations (1)

(ii) $P(\sqrt{x}) = 4(\sqrt{x})^3 + 8(\sqrt{x})^2 - 1 = 0$ (1)

$\therefore 4x^{3/2} + 8x - 1 = 0$

$4x^{3/2} = -8x + 1$ (1)

$\therefore 16x^3 = 64x^2 - 16x + 1$ *

$\therefore 16x^3 - 64x^2 + 16x - 1 = 0$ (1)

(iii) $\frac{1}{\alpha^4} + \frac{1}{\beta^4} + \frac{1}{\gamma^4} = \frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha^2\beta^2\gamma^2}$ (1)

$= \frac{\text{Sum of roots} \times 2}{\text{Prod. of roots}}$ from equation *

$= \frac{1/16}{1} = 16$ (1)

(iv) $OA^2 = \alpha^2 + \left(\frac{1}{\alpha^2}\right)^2 = \alpha^2 + \frac{1}{\alpha^4}$

$OB^2 = \beta^2 + \frac{1}{\beta^4}$

$OC^2 = \gamma^2 + \frac{1}{\gamma^4}$

$\therefore OA^2 + OB^2 + OC^2 = \alpha^2 + \beta^2 + \gamma^2 + \frac{1}{\alpha^4} + \frac{1}{\beta^4} + \frac{1}{\gamma^4}$

$= 4 + \frac{256}{2}$

$= 132$

$\frac{1}{\alpha^4} + \frac{1}{\beta^4} + \frac{1}{\gamma^4} = \frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha^2\beta^2\gamma^2}$

and $\alpha^4\beta^4 + \alpha^4\gamma^4 + \beta^4\gamma^4 = (\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2)^2 - 2\alpha^2\beta^2\gamma^2(\alpha^2 + \beta^2 + \gamma^2)$

$= 1^2 - 2 \times \frac{1}{16} \times 4$

$= \frac{1}{2}$

$\alpha^4\beta^4\gamma^4 = \frac{1}{256}$

QUESTION 6

a) i) $x = a \cos \theta$

$$\frac{dx}{d\theta} = -a \sin \theta$$

$$y = b \sin \theta$$

$$\frac{dy}{d\theta} = b \cos \theta$$

$$\therefore \frac{dy}{dx} = \frac{b \cos \theta}{-a \sin \theta}$$

Eqn of tangent.

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$\frac{y \sin \theta}{b} - \sin^2 \theta = -\frac{x \cos \theta}{a} + \cos^2 \theta$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = \sin^2 \theta + \cos^2 \theta$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

ii) $N = (a \cos \theta, 0)$

when $y = 0$

$$x = \frac{a}{\cos \theta}$$

$$\therefore T = \left(\frac{a}{\cos \theta}, 0 \right)$$

$$ON \cdot OT = a \cos \theta \cdot \frac{a}{\cos \theta} = a^2$$

iii

$$P (a \cos \theta, b \sin \theta)$$

$$S (ae, 0)$$

$$R \left(\frac{a}{e}, \frac{b(e - \cos \theta)}{e \sin \theta} \right)$$

$$M_{PS} = \frac{b \sin \theta}{a \cos \theta - ae}$$

$$\frac{e \sin \theta}{\frac{a}{e} - ae}$$

$$= \frac{be - b \cos \theta}{e \sin \theta} \times \frac{e}{a - ae^2}$$

$$= \frac{b(e - \cos \theta)}{a(1 - e^2) \sin \theta}$$

$$M_{PS} \times M_{SR}$$

$$= \frac{b \sin \theta}{a \cos \theta - ae} \times \frac{b(e - \cos \theta)}{a(1 - e^2) \sin \theta}$$

$$= \frac{-b^2}{a^2(1 - e^2)}$$

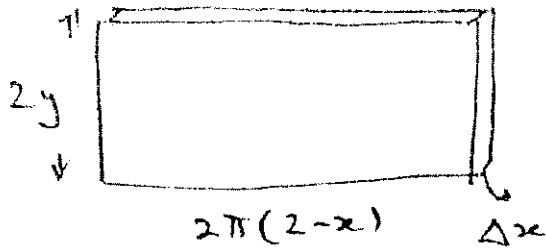
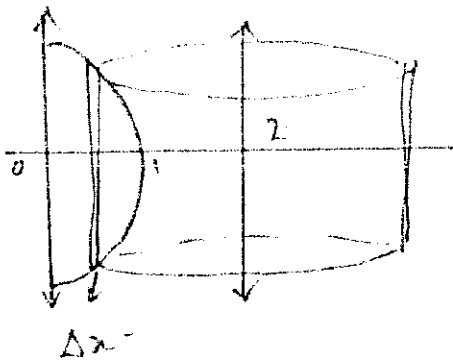
but $b^2 = a^2(1 - e^2)$ for ellipse

$$= -1$$

$$\therefore \angle PSR = 90^\circ$$

QUESTION 6. (cont)

b. ii)



$$\Delta V = 2\pi(2-x) \cdot 2y \cdot \Delta x$$

$$\begin{aligned} \text{Volume} &= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^1 2\pi(2-x) \cdot 2\sqrt{1-x} \Delta x \\ &= 16\pi \int_0^1 (2-x)\sqrt{1-x} dx \end{aligned}$$

ii) Let $u = 1-x$

$\therefore du = -dx$

$$\text{Volume} = 16\pi \int_1^0 ((1+u)\sqrt{u}) \cdot -du$$

$$= 16\pi \int_0^1 (\sqrt{u} + u\sqrt{u}) du$$

$$= 16\pi \left[\frac{2}{3} u^{3/2} + \frac{2}{5} u^{5/2} \right]_0^1$$

$$= \frac{256\pi}{15}$$

QUESTION 7

a. i) roots of $z^3 = 1$ are

1

$$w_1 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$$

$$w_2 = \cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3}$$

$$\begin{aligned}\therefore \overline{w_2} &= \overline{\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3}} \\ &= \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \\ &= w_1\end{aligned}$$

$$\begin{aligned}w_2^2 &= \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3}\right)^2 \\ &= \cos \frac{4\pi}{3} - i \sin \frac{4\pi}{3} \\ &= \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \\ &= w_1\end{aligned}$$

ii) as $1, w_1, w_2$ are the roots of $z^3 - 1 = 0$

$\therefore$ Sum of roots $\left(-\frac{b}{a}\right)$

$$1 + w_1 + w_2 = 0$$

$$w_1 + w_2 = -1$$

iii) product of roots $\left(\frac{c}{a}\right)$

$$\therefore 1 \times w_1 \times w_2 = 1$$

$$w_1 w_2 = 1$$

b. i) integral represents the area of $\frac{1}{4}$ of a circle radius a ,

$$\therefore \int_0^a \sqrt{a^2 - x^2} dx = \frac{1}{4} \pi a^2$$

$$\begin{aligned}\text{or } \int_0^a \sqrt{a^2 - x^2} dx & \quad x = a \sin \theta \\ dx &= a \cos \theta d\theta\end{aligned}$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{a^2 - a^2 \sin^2 \theta} \cdot a \cos \theta d\theta$$

$$= a^2 \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$= \frac{a^2}{2} \int_0^{\frac{\pi}{2}} 1 + \cos 2\theta d\theta$$

$$= \frac{a^2}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{4} \pi a^2$$

$$\text{ii) } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\therefore y^2 = \frac{b^2}{a^2} (a^2 - x^2)$$

$$\therefore y = \frac{b}{a} \sqrt{a^2 - x^2}$$

$$\therefore \text{area} = 4 \times \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$$

$$= \frac{4b}{a} \times \frac{1}{4} \pi a^2$$

$$= \pi ab$$

iii) area of cross-section

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \lambda^2$$

$$\text{or } \frac{x^2}{a^2 \lambda^2} + \frac{y^2}{b^2 \lambda^2} = 1$$

is given by $A = \pi ab \lambda^2$

Q7(cont)

but $z^2 = 1 - \frac{h^2}{H^2}$

$$\therefore A = \pi ab \left(1 - \frac{h^2}{H^2}\right)^2$$

$$\begin{aligned} \therefore V &= \pi ab \int_0^H \left(1 - \frac{h^2}{H^2}\right)^2 dh \\ &= \pi ab \int_0^H \left(1 - \frac{2h^2}{H^2} + \frac{h^4}{H^4}\right) dh \\ &= \pi ab \left[h - \frac{2h^3}{3H^2} + \frac{h^5}{5H^4} \right]_0^H \\ &= \pi ab \times \frac{8H}{15} \\ &= \frac{8\pi ab H}{15} \end{aligned}$$

QUESTION 8

a i) $y = c^2 x^{-1}$
 $\frac{dy}{dx} = -c^2 x^{-2}$

when $x = cp$

$$\begin{aligned} m_T &= \frac{-c^2}{c^2 p^2} \\ &= -\frac{1}{p^2} \end{aligned}$$

$$\therefore y - \frac{c}{p} = -\frac{1}{p^2} (x - cp)$$

$$p^2 y - cp = -x + cp$$

$$x + p^2 y = 2cp$$

ii) Solve simultaneously to find (x_1, y_1)

$$x + p^2 y = 2cp$$

$$x + q^2 y = 2cq$$

subtract

$$p^2 y - q^2 y = 2cp - 2cq$$

$$y = \frac{2c(p-q)}{p^2 - q^2}$$

$$= \frac{2c}{p+q}$$

$$\therefore p+q = \frac{2c}{y_1} \quad (B)$$

$$\therefore x + p^2 \left(\frac{2c}{p+q} \right) = 2cp$$

$$x = 2cp - \frac{2cp}{p+q}$$

$$= \frac{2cp(p+q) - 2cp}{p+q}$$

$$= \frac{2cpq}{p+q}$$

$$\therefore \frac{x_1}{y_1} = \frac{\frac{2cpq}{p+q}}{\frac{2c}{p+q}}$$

$$= pq \quad (A)$$

iii) $d^2 = (cp - cq)^2 + \left(\frac{c}{p} - \frac{c}{q}\right)^2$
 by distance formula

$$= c^2 \left[(p-q)^2 + \frac{(q-p)^2}{p^2 q^2} \right]$$

$$= c^2 \left[(p-q)^2 \left(1 + \frac{1}{p^2 q^2}\right) \right]$$

iv) $d^2 = c^2 (p-q)^2 \left(1 + \frac{1}{p^2 q^2}\right)$

sub: $\frac{x}{y} = pq$

$$\frac{2c}{y} = p+q$$

$$(p-q)^2 = (p+q)^2 - 4pq$$

$$\therefore d^2 = c^2 \left(\frac{4c^2}{y^2} - \frac{4x}{y} \right) \left(1 + \frac{y^2}{x^2} \right)$$

$$d^2 = 4c^2 \left(\frac{c^2 - xy}{y^2} \right) \left(\frac{x^2 + y^2}{x^2} \right)$$

$$x^2 y^2 d^2 = 4c^2 (c^2 - xy)(x^2 + y^2)$$

$$b. i) f(x) = e^{-x} \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} \right) \quad x \geq 0$$

$$\begin{aligned} f'(x) &= -e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) + e^{-x} \left(1 + \frac{x}{1} + \frac{x^2}{2!} + \dots + \frac{x^{n-1}}{(n-1)!} \right) \\ &= -e^{-x} \left(\frac{x^n}{n!} \right) \end{aligned}$$

$$< 0 \quad \text{for all } x \geq 0$$

$$\text{as } e^{-x} > 0$$

$$\frac{x^n}{n!} > 0$$

$\therefore$ curve is decreasing for all $x \geq 0$

$$\text{as } f'(x) < 0$$

ii) when $x < 0$ $f(0) = 1$

also curve is always decreasing

$$\therefore f(x) \leq 0$$

$$\therefore e^{-x} \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} \right) \leq 1$$

$$\therefore e^x \geq 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$$